DKT Data Preprocessing - Milestone Report

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1. Purpose

This report describes how the raw student data was cleaned and prepared for the Deep Knowledge Tracing (DKT) model. The DKT model will read a student's answer history and predict the chance that the student answers each skill correctly next time. These predictions will be used by the prerequisite planner of the tutor.

This data is frozen. After this milestone is submitted, the processed data and the steps below will not be changed.

2. Dataset

- **Name:** ASSISTments 2009-2010 skill builder data (school maths practice data)
- **File used:** skill_builder_data.csv, from the ASSISTments 2009-2010 skill-builder page
- **Size:** 303,260 rows
- **License:** No formal license was found on the official data page. The page asks users to cite the data page and the paper below. The data is used for academic research only and is not re-shared.
- **Citation:** Feng, M., Heffernan, N., and Koedinger, K. (2009). Addressing the assessment challenge with an online system that tutors as it assesses.

3. Columns used

Only 4 columns are used. The other columns (for example teacher, school and class information) do not describe what the student knows, and the real tutor will not have them for a new student.

Column	Meaning
user_id	Which student
order_id	Time order of the answers

Column	Meaning
skill_id	Which skill (topic) the question tested
correct	1 = right on the first try, 0 = wrong

4. Cleaning steps

Step	What was done	Rows after step
1	Loaded the 4 columns	303,260
2	Removed 1 row with missing values	303,259
3	Removed 46,668 duplicate rows (same student, order_id and skill)	256,591
4	Removed 325 students who have fewer than 3 steps	256,073

After step 3 there are 3,733 students, 82 skills, and 65.5% correct answers. After step 4 there are 3,408 students.

5. Encoding

- Skill IDs were renumbered from 0 to 81 (N = 82 skills).
- Every step becomes one number: **x = skill + 82 x correct**.
- x goes from 0 to 163, so the model input has 164 possible values.
- Example: skill 7 answered wrong gives 7. Skill 7 answered right gives 89.

6. Sequences

- Rows were sorted by user_id and then order_id, so each student has a list in time order.
- Many questions have more than one skill. These rows were kept as separate steps, as most DKT papers do.
- Some students have very long histories (the longest has 6,149 steps). Cutting every student at 200 steps would remove 70,414 rows (about 27% of the data). So long students were split into **chunks of 200 steps**, and no data was lost.

- Chunks with fewer than 3 steps were dropped (8 chunks, 13 steps in total).
- Shorter sequences were padded with zeros to length 200. A **mask** marks real steps (1) and padding (0), so the model ignores the padding.

7. Train and test split

- The split is by **student**, not by row. All chunks of one student stay in the same set, so the test students are completely new to the model.
- Split: 80% train and 20% test, random seed 42.

Set	Students	Sequences (shape)	Real steps	Correct rate
Train	2,726	3,168 x 200	205,418	0.657
Test	682	781 x 200	50,642	0.644
Total	3,408	3,949	256,060	-

8. Output files

The file **dkt_data.npz** holds these arrays for train and test:

- **X**: input id of each step
- **S**: skill of each step
- **C**: correct (0 or 1) of each step
- **M**: mask (1 = real step, 0 = padding)
- **N**: number of skills (82)

9. Settings used

Setting	Value
Maximum sequence length	200
Minimum steps per student	3
Train / test split	80% / 20% by student
Random seed	42

Setting	Value
Number of skills	82

10. Checks that passed

- Row counts match: 303,260 - 1 - 46,668 = 256,591.
- The encoding formula x = skill + 82 x correct is true for every real step.
- Train and test have the expected number of students (2,726 and 682).
- The correct rate is similar in train (0.657) and test (0.644).
- Total real steps (256,060) are close to the expected 256,073. The difference of 13 steps is the dropped short chunks.

11. Limitations

- A chunk after the first one starts without the student's earlier history.
- Questions with many skills are counted as many steps.
- The test set has a slightly lower correct rate than the train set. The students are different, so this is normal.
- The skills come from the ASSISTments data. They must be mapped by hand to NCERT skills for the tutor.